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MILITARY SPECIFICATION

# CLOSURE FOR ROCKET, HE, 66MM, ANTITANK: M72A2

This specification is approved for use by all Activities of the Department of the Army.

## 1. SCOPE

1.1 This specification contains requirements not normally required by the drawings, and provides quality assurance provisions for the fabrication and packing of one type of closure for Rocket, HE, 66MM Anti-Tank Weapon, M72A2.

## 2. APPLICABLE DOCUMENTS

2.1 The following documentsof the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-A-2550 | Ammunition and Special Weapons; General Specification for |
| MIL-A-8625 | Anodic Coatings for Aluminum and Aluminum Alloys |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-1168 | Lot Numbering of Ammunition |

FSC 1340

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| MIL-STD-1169 | Packaging, Packing and Marking for Shipment of Inert Ammunition Components |
| MIL-STD-1235 | Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

DRAWINGS

Army

| | |
|---|---|
| 9220251 | Closure |
| 9246403 | Calibration Standards |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. The following document form part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| ASTM-B117 | Salt Spray (Fog) Testing, |
| ASTM-D2290 | Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method Standard Method of |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

## 3. REQUIREMENTS

3.1 Material. Materials used in the fabrication of the closure shall be in accordance with applicable drawings and specifications.



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3.2 Closures. The closures shall comply with all requirements specified on Drawing (dwg) 9220251, and with all requirements specified in applicable specifications.

3.3 Heat treatment

3.3.1 Maximum heat-treatment temperatures. The material used in the fabrication of the closure shall at no time be subjected to Fahrenheit temperatures in excess of 880 degrees for 7001 alloy and 900 degrees for ZG93 alloy. If a salt bath furnace is utilized for solution heat-treatment, the cold-worked closures shall be pre-heated to relieve internal stresses to preclude the development of cracks during the normal rapid rise in temperature upon introduction into the salt bath.

3.3.2 Grain structure. The closure, after all heat-treatments and aging treatments have been completed, shall not contain eutectic melting, defective grain boundary conditions or banding (segregation of inter-metallic compounds) when examined under 50 power magnification.

3.4 Physical strength

3.4.1 Yield strength. Prior to application of protective coating, the pitch diameter of the threads and the threaded cavity of the closure shall not be deformed, nor shall the closure rupture while being subjected to an internal hydrostatic pressure of 14,000 plus or minus 200 pounds per square inch, while being supported only by the threads with all other surfaces of the closure in a non-confined, non-restricted condition.

3.4.2 Ultimate strength. Prior to application of protective coating, the pitch diameter of the threads shall maintain assembly and the closure shall not rupture while being subjected to an internal hydrostatic pressure of 14,800 pounds per square inch, while being supported only by the threads with all other surfaces of the closure in a non-confined, non-restricted condition.

3.5 Protective coating. The exterior and interior surfaces of the closure, after complete application of the protective coating, shall not contain in excess of four (4) total isolated bare spots or pits larger than one thirty-second (1/32) of an inch in diameter after the finished closure has been exposed to a salt spray solution containing five (5) percent salt for a minimum of one hundred sixty eight (168) hours.



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3.6 Physical defects. The completely finished closure shall be free from all external and internal cracks, metallic discontinuities, laps and surface seams when the entire exterior surface is scanned by ultrasonic inspection equipment adjusted to the sensitivity to detect defects of the magnitude described and depicted in dwg. 9246403. The entire pressure cavity wall and base, including the bulkhead and threads, shall also be free from these defects. Seven (7) power magnification shall be utilized to determine the presence of surface defects in those areas which are not accessible to the ultrasonic probe.

3.7 Workmanship. All closures shall be free of chips, dirt, grease, rust, scale and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All markings shall be clean and sharply defined.

3.8 Submission of first article sample. This specification makes provisions for the submission of a first article sample for evaluation by the Government. Procedures for submission of the first article samples shall be as specified in the contract (See 6.4).

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:



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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principle. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission. Prior to initiating regular production, the contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2 (see 6.4). The first article sample shall consist of:

20 closures that have not been subjected to hydrotesting or anodizing.

15 closures that are completely finished and represent product intended for delivery.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a



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lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspection to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3.1 of this specification and any or all of the requirements of the applicable drawing.

4.2.3 Rejection. The first article sample shall be rejected if any component or assembly fails to comply with any of the applicable requirements referenced in 4.3.2 and 4.3.3.1. The first article sample shall also be rejected if one or more specimens are classed defective during the ultimate-strength test of motor bodies (see 4.3.3.1.2) or if the average burst or failure level of the combined sample is less than 16,500 pounds per square inch. The Government reserves the right to terminate its inspection upon observing any non-conformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Quality Conformance (Production Lot Acceptance) Inspection

4.3.1 Lot formation. The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer,



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in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects shall be performed 100 percent. Certain Major Defects, when so specified, shall be examined 100 percent. Sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of. 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph, except when 100 percent inspection is specified. This option shall be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Closure, after hydrostatic test and after anodizing (see 3.4.1, 3.5 and dwg. 9220251).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | | | |
| 1. | Hydrostatic test stamp missing (see 6.5.1) ..... | Visual | 01001* |
| 2. | Physical defects test stamp missiong (see 6.5.1)..... | Visual | 01002* |



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| | | | |
|---|---|---|---|
| Major: | AQL 1.50 Percent | | |
| 101. | Length of thread, minimum (min.)..... | Gage | 01003 |
| 102. | Diameter of counterbore at small end..... | Gage | 01004 |
| 103. | Runout of minor diameter with small inside diameter..... | Gage | 01005 |
| 104. | Length to bottom of threaded cavity from small end..... | Gage | 01006 |
| 105. | Depthof smallest diameter cavity, min..... | Gage | 01007 |
| 106. | True position of large diameter with small diameter..... | Gage | 01008 |
| 107. | Pitch diameter of thread, maximum (max.)..... | Gage | 01009 |
| 108. | Diameter of smallest cavity..... | Gage | 01010 |
| 109. | Weight, max..... | Gage | 01011 |
| 110. | Small outside diameter..... | Gage | 01012 |
| 111. | Outside shoulder radius, Min..... | Gage | 01013 |
| 112. | Protective coating damaged, missing or incomplete..... | Visual | 01014 |
| Minor: | AQL 1.50 Percent | | |
| 201. | Total length..... | Gage | 01015 |
| 202. | Depth of smallest diameter cavity, max. ..... | Gage | 01016 |
| 203. | Minor diameter of thread, max. ..... | Gage | 01017 |
| 204. | Evidence of poor workmanship (see 3.7)..... | Visual | 01018 |

4.3.3 Testing

4.3.3.1 First article sample. The first article sample shall be examined 100 percent for all of the characteristics in 4.3.2, (without regard to classification and AQL) and then subjected to the tests listed below. Inspection and test equipment shall conform to the requirements of 4.3.4. Rejection of the first article sample shall be in accordance with 4.2.3.

4.3.3.1.1 Yield strength (see 3.4.1) Code No. 01019. The twenty (20) unfinished (not hydrotested or anodized) closures shall be tested as specified in 4.4.1.



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4.3.3.1.2 Ultimate strength (see 3.4.2) Code No. 01020. Seventeen (17) of the closures from 4.3.3.1.1 shall be tested as specified in 4.4.2. Samples subjected to this test shall be scrapped.

4.3.3.1.3 Hoop tensile strength (see dwg. 9220251) Code No. 01021. Three (3) of the closures from 4.3.3.1.1 shall be tested as specified in 4.4.3.

4.3.3.1.4 Physical defects (see 3.6) Code No. 01022. Ten (10) of the finished closures shall be tested as specified in 4.4.4.

4.3.3.1.5 Surface finish (see dwg. 9220251) Code No. 01023. The ten (10) closures from 4.3.3.1.4 shall be tested as specified in 4.4.5.

4.3.3.1.6 Protective coating

4.3.3.1.6.1 Salt spray method (see 3.5) Code No. 01025. Six (6) of the closures from 4.3.3.1.5 shall be tested as specified in 4.4.6.

4.3.3.1.6.2 Direct Measurement (See dwg. 9220251) Code No. 01026. Four (4) of the closures from 4.3.3.1.5 shall be tested as specified in 4.4.7.

4.3.3.1.7 Grain structure and direction. (See 3.3.2 and dwg. 9220251). Defective grain structure; Code No. 01027; Improper grain direction, Code No. 01028. Five (5) of the finished closures shall be tested as specified in 4.4.8.

4.3.3.2 Production Sample.

4.3.3.2.1 Yield strength (see 3.4.1) critical defect, Code No. 01019*. This test shall be performed one hundred percent (100%). The test shall be performed as specified in 4.4.1. Defective closures shall be rejected and removed from the lot.

4.3.3.2.2 Ultimate strength (see 3.4.2) Code No. 01020. This test is to be conducted FOR INFORMATION ONLY. A sample of twenty (20) closures shall be selected from each closure inspection lot (after satisfactory conformance to 4.3.3.2.1) and tested as specified in 4.4.2. The test results shall be recorded and copies furnished as specified by the contract (see 6.1). Samples subjected to this test shall be scrapped.



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4.3.3.2.3 Hoop tensile strength (see dwg. 9220251) Major defect, Code No. 01021. A sample of five (5) closures shall be selected from each heat-treated batch (prior to anodizing) from near the center of the group of the closures. The samples shall be tested as specified in 4.4.3. The heat-treatment batch shall be rejected if the average hoop tensile strength is less than the specified drawing requirement.

4.3.3.2.4 Physical defects (see 3.6) critical defect, Code No. 01022*. This test shall be performed one hundred percent (100%). The test shall be performed as specified in 4.4.4. Defective closures shall be rejected and removed from the lot.

4.3.3.2.5 Surface finish (see dwg. 9220251) major defect, Code No. 01023. A sample of ten (10) completely finished closures shall be selected from the lot and tested as specified in 4.4.5. The lot shall be rejected if two or more closures are classed defective.

4.3.3.2.6 Protective against atmospheric stress corrosion (see dwg. 9220251) critical defect, Code No. 01024*. The contractor shall establish process controls and schedules to faciliate the Government's surveillance of the contractor's conformance to assure not exceeding the allowable elapsed time from when any surface of the closure is final machined until the protective coating is applied. All products not adquately protected shall be rejected, removed from processing and scrapped.

4.3.3.2.7 Protective coating. In determining the adquacy of the protective coating, the contractor may select either of the following methods for acceptance testing of regular production of closures for the Government.

4.3.3.2.7.1 Salt spray method (see 3.5) major defect, Code No. 01025. A sample of three (3) closures shall be selected from each inspection lot and tested as specified in 4.4.6. The inspection lot shall be rejected if one or more closures in the sample are classed defective.

4.3.3.2.7.2 Direct measurement method (see dwg. 9220251) major defect, code No. 01026. A sample of three (3) closures shall be selected from each inspection lot and tested as specified in 4.4. The inspection lot shall be rejected if one or more closures in the sample are classed defective.



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4.3.3.2.8 Grain structure, grain direction and maximum heat-treatment temperature. Critical defects; defective grain structure (see 3.3.2), Code No. 01027*; Improper Grain Direction (see dwg. 9220251), Code No. 01028*; and Excessive Heat-Treatment Temperature (see 3.3.1), Code No. 01029*. A sample of five (5) closures, after all heat-treating and aging treatments, shall be selected from each heat/aging treatment batch and tested as specified in 4.4.8. The batch shall be rejected if one or more closures are classed defective, or if the temperature recording equipment indicates that the upper temperature limit had been exceeded during any phase of fabrication.

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests required by 4.3.2 and 4.3.3 of this specification shall be designed and fabricated by the contractor. Refer to 6.3 for details concerning responsibilities for inspection equipment design and Government approval.

4.4. Test methods and procedures. The test methods and procedures described in each of the following tests represent only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using these or substitute methods and procedures. Whichever option is selected, detailed descriptions must be submitted to the Government for approval prior to use. Government approval will be based on an evaluation and demonstration to determine the effectiveness of all of the tests and test procedures.

4.4.1 Yield strength of closure. The closure, after all machining operations have been completed, and prior to anodizing, shall be assembled to the applicable test equipment and subjected to the internal hydrostatic pressure specified for a minimum of 2 seconds. The medium used in testing may be water, or any liquid compatible with the closure and test equipment. The test equipment shall automatically apply hydrostatic pressures within the prescribed limits. The test equipment shall automatically stamp the component if the component has actually been subjected to the required pressures. The test equipment shall automatically assure that the component is not stamped if it is not subjected to the minimum level specified. The test equipment shall have automatic controls that will lock the equipment in an inoperative condition at any time that the upper limit of test pressure is exceeded. Components subjected to pressures that exceed the upper specified limit shall be rejected and not returned to any lot. Repair and adjustments to the test equipment shall follow any occurrence of excessive pressure. After testing, the closure shall be dried and visually examined for bulges and deformation. Closures that appear to be deformed beyond drawing dimensions, burst or have threads that fail to maintain assembly shall be classed defective.



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4.4.2 Ultimate strength of closure. The closure, after having exhibited acceptable yield strength characteristics, shall be assembled to the applicable destructive test equipment and subjected to an internal hydrostatic pressure of increasing magnitude until the closure ruptures or until the threads pull out from the test fixture. The pressure level, when failure occurs, shall be recorded to the nearest fifty (50) pounds per square inch and the results distributed in accordance with the instructions in 6.1.

4.4.3 Hoop tensile strength. The closure shall be sectioned, transverse to the longitudinal axis, to produce a ring 0.10 inch plus 0.01 inch in length from between 7/8 inch and 1 inch from the unthreaded end of the closure. The ends of the ring shall be parallel within 0.002 inch. The length of the finished specimen shall be measured to the nearest 0.001 inch and the result recorded as "l". The wall thickness shall be measured at a minimum of four (4) equally spaced locations, the measurements averaged to the nearest 0.001 inch and the result recorded as "t". The specimen shall be assembled to the split disk fixture described in the Standard Method of Test for Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method: Designation D2290 of the ASTM Standards. The load will be applied at the rate specified in D2290 until failure occurs. The hoop tensile strength shall be calculated using the following formula:

$$S = \frac{L}{2A}$$

Where:

S = the hoop tensile strength in pounds per square inch
L = load at failure in pounds
A = Cross-sectioned area in square inches (the product of "l" & "t")

4.4.4 Physical defects. Following all manufacturing, inspection, other testing and protective coating application operations, the closure shall be placed on the specified test equipment and the smaller diameter (for its entire length;) the entire pressure cavity wall and base, including the threads, shall be scanned using the applicable ultrasonic inspection equipment. Areas not accessible to ultrasonic scanning shall be examined visually using the magnification specified. Standard closures containing designed defects (see dwg. 9246403) shall be utilized to calibrate the equipment to produce



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consistent signals of the magnitude specified in the equipment operating and calibration procedure. The test standard closures shall then be tested for a consistent rejection signal. Acceptable closures only shall be stamped with proper marking indicating the closure has been subjected to the test and was not rejected. Closures containing defects which produce signal indications equal to or in excess of the gate level specified in the operating and calibration procedure, or closures containing visual defects in areas not covered by the ultrasonic equipment shall be classed defective and removed from the lot. All closure stamped since the last successful equipment calibration verification shall be rejected, subject to retest by the contractor witnessed by the Government Inspector; if during any verification, the test equipment fails to reject the test standards or if the test standards fail to reproduce a signal of the specified magnitude or if the gate level fails to be operating at the specified level.

4.4.5 Surface finish. The surface finish of the smaller inside diameter may be determined by comparison to the finish blocks of American Standard USASB46.1 referenced in MIL-A-2550 or measured directly by profilometer or other commercial equipment in accordance with 4.3.4. The closure shall be classed defective if the finish fails to conform to the drawing requirement.

4.4.6 Protective coating (salt spray method). The component shall be tested as prescribed in Standard Method B117 in the "American Society for Testing and Materials" (ASTM) using the required salt concentrations for the required period of time. After testing, the components may be rinsed in warm water, without rubbing, allowed to dry, without wiping, and visually examined. The component shall be classed defective if bare spots or the visible corrosion exceeds the specified limits.

4.4.7 Protective coating thickness (direct measurement). The closure shall be sectioned to yield two halves. The primary surfaces of interest are diametrically opposed longitudinal sections containing longitudinal exposure of the radius at the bottom of the threaded cavity. Both edges of one specimen shall be polished and etched to reveal the protective coating thickness. This thickness shall be measured to the nearest 0.00005 inch. The component shall be classed defective if protective coating thickness fails to conform to the minimum thickness required by the applicable drawing.

4.4.8 Grain structure and direction. The closure shall be sectioned transverse to the longitudinal axis approximately one-eighth (1/8) inch from the threaded end. The newly exposed circular surface of the 1/8 inch long portion shall be polished, etched and visually examined under the specified magnification to determine



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condition of the grain structure. The remaining portion shall be sectioned along the longitudinal centerline (approximately) and both edges of one half shall be polished and etched to determine condition of grain. Observation in the area of juncture of the bulkhead (barrier, partition) with the cylindrical edge of the closure shall be made to determine grain direction. The closure shall be classed defective if any portion of any specimen fails to comply with the specified requirement.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking - level C. The closures shall be packaged, packed and marked in accordance with MIL-STD-1169.

## 6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

c. Instructions for distribution of results of ultimate strength tests of closures, and results of first article static firing tests of complete assemblies. Results shall be forwarded to MICOM, AMSMI-US and ARMCOM, SARPA-QA-A-R in accordance with appropriate Data Item Description per DODI 5010.12 as directed by the contracting officer.

d. Provisions for submission of first article sample.

6.2 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the Inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.3).

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the Method of Inspection headings in Section 4). Design responsibility for all other



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inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1 c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.



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6.3.3 Submission of Designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirments List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of first article sample. Instructions as to the location for the evaluation of first article sample shall be obtained from the Contracting Officer. Delivery of test data, hardware and inspection records shall be in accordance with appropriate contract documents.

6.5 Submission of Critical Defect Reports. For U. S. Army procurements, one copy of each Critical Defect report will be furnished the cognizant Government inspection agency.

6.5.1 Critical defect reports. The contractor's quality assurance system shall contain a procedure for reporting of Critical defectives found during the course of his inspection and during inspection by the Government. Any Critical Defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those Critical Defects annotated by an asterisk will be furnished. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.

b. Number and revision date of specification.

c. Characteristic code number.

d. Lot or serial number.

e. Number of defectives found.

f. Assignable cause of analysis or occurrence.

g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.



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6.5.2 Reporting to other services. For other than U. S. Army procurement, Critical Defect reporting shall be as prescribed by the Contracting Officer.

6.6 The following information is listed herein as advisory to assist the manufacturer in the production of closures.

6.6.1 Material. The extrusion slug and finished closure should be free from surface seams, laps, internal cracks, porosity or other metallic discontinuities including excess segregations. These conditions are readily visible when the end of the extrusion slug or section of a finished closure is micro-etched and examined visually. When sections of the closure material and examined under 50 power magnification, the cross section should show no evidence of eutectic melting, excessive high temperature oxidation or other harmful grain boundary conditions and longitudinal sections should show no evidence of banding (zebra effects caused by segregation of intermetallic compounds).

6.6.2 Extruded slug. Care should be exercised to avoid sharp corners and high stress concentration areas, particularly at the base of the pressure cavity area, during the extrusion of the pre-machined part.

6.6.3 Heat treatment. The extrusion slugs should be in the fully annealed condition prior to extruding or forming. The slugs should never be subjected to temperatures in excess of 880 degrees Fahrenheit for 7001 alloy and 900 degrees Fahrenheit for ZG93 alloy.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

Custodian:
Army - MI

Review Activity:
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Preparing Activity:
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